

Unique Factorization Domains

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We will show that not every PID is a Euclidean domain by relating the principal ideal property with a weakening of the Euclidean condition.

Definition 1. Let N be a **Dedekind-Hasse norm** if $N : R \rightarrow \mathbb{Z}^{\geq 0}$ satisfies the following conditions:

1. $N(a) = 0$ if and only if $a = 0_R$.
2. For every nonzero $a, b \in R$, either $a \in (b)$ or there's a non-zero element $x \in (a, b)$ such that $N(x) < N(b)$.

The second condition can be rephrased as: either a divides b or there exists $s, t \in R$ such that $N(as - bt) < N(b)$.

Notice that R is a Euclidean domain if we can pick N such that $s = 1$ and this condition is satisfied.

Proposition 0.1. An integral domain R is a PID if and only if R has a Dedekind-Hasse norm.

Proof. Assume R has a Dedekind-Hasse norm.

Let I be a non-zero ideal in R and let $b \in I$ be such that $N(b)$ is minimal. We will show that $I = (b)$.

Let $a \in I$. If $a = 0$, we're done.

Assume that $0 \neq a \notin (b)$ by contradiction. Then, by the Dedekind-Hasse condition, there exists $x \in (a, b)$ such that $N(x) < N(b)$. But $(a, b) \subseteq I$, so we have contradicted the minimality of $N(b)$.

Now, assume that R is a PID. Then, R is also a UFD.

Define N by letting $N(0) = 0$, $N(u) = 1$ if u is a unit, and $N(a) = 2^n$ if $a = p_1 \dots p_n$ is the irreducible decomposition of a .

Clearly, N is multiplicative. Let's now show that N is a Dedekind-Hasse norm.

□

Let $R = \mathbb{R}[x, y]/(x^2 + y^2 - 1)$ be the coordinate ring of the unit sphere over the reals. Let $X = V(x^2 + y^2 - 1)$.

Intuitively, $x^2 + y^2 - 1$ is irreducible as the circle would be a union of two lines otherwise, but it clearly is not.

Rigorously, $x^2 + y^2 - 1$ is irreducible as

We will show that R is not a UFD by showing that $\bar{x}$ is irreducible but not prime.

Notice that $V(\bar{x})$ consists of two points, the north and south poles of the sphere.

Thus, $\bar{x}$ is not zero since otherwise $V(\bar{x}) = X$.

Similarly, $\bar{y}$ is not a unit since otherwise $V(\bar{y}) = \emptyset$.

The statement that $\bar{x}$ is not prime simply corresponds to the geometric fact that $V(\bar{x})$ is reducible, as it's a union of two points.

Lemma 0.2. $\bar{x}$ is not prime.

Proof. We will show that $\bar{x}$ doesn't divide $1 \pm \bar{y}$. Since $\bar{x}^2 = (1 + \bar{y})(1 - \bar{y})$, this shows that $\bar{x}$ is not prime since $(1 + \bar{y})(1 - \bar{y})$ is in the ideal $(\bar{x})$ but neither factor is in $(\bar{x})$.

Assume $\bar{x}$ divides $1 + \bar{y}$.

Then, $1 + y = xg + (x^2 + y^2 - 1)h$ for some $g, h \in \mathbb{R}[x, y]$.

Plugging in $(x, y) = (0, 1)$ produces $2 = 0$, which is a contradiction. □

Lemma 0.3. $\bar{x}$ is irreducible.

Proof. □

Notice what the issue is here. $\bar{x}$ can't be written as a product of two functions, but the ideal $(\bar{x})$ is the intersection of the ideals $I(\bar{x}, \bar{y} - 1)$ and $I(\bar{x}, \bar{y} + 1)$ using the Nullstellensatz and that intersections of ideals correspond to unions of the varieties.

Let R be a UFD. Is there a geometric condition that captures this?

Are localizations of R also UFDs?